

BioTime Migration Guide

Migration from ISPADMIN2 to IBN-SERVER

Item	Value
Old live machine	ISPADMIN2
New server	IBN-SERVER
BioTime URL on new server	http://192.168.64.4:8081
Database name	biotime
SQL login	ISPAdmin
Migration method	Schema Compare + Data Compare, because backup/restore is blocked by SQL Server version difference

Important: During final migration, do not let ISPADMIN2 and IBN-SERVER both receive live attendance punches at the same time. Stop the old BioTime system first, then sync schema/data, then start the new server.

Before Migration

1. Confirm BioTime opens on IBN-SERVER at `http://192.168.64.4:8081`.
2. Confirm these services are running on IBN-SERVER:

```
bio-redis    Running
bio-cache    Running
bio-monitor   Running
```

3. Confirm the SQL login exists and is enabled on IBN-SERVER:

```
USE master;
GO

SELECT name, is_disabled
FROM sys.sql_logins
WHERE name = 'ISPAdmin';
GO
```

If the login is disabled:

```
USE master;
GO

ALTER LOGIN ISPAdmin ENABLE;
GO
```

4. Keep a rollback copy of the current IBN-SERVER database if needed. If the current database is only a test database, rename it before final sync:

```
ALTER DATABASE biotime MODIFY NAME = biotime_before_migration;
GO
```

Final Migration Steps

1. **Choose a quiet time.** Do the final migration when no employees/devices are actively sending attendance punches.
2. **Stop BioTime services on ISPADMIN2.** Open Command Prompt as Administrator on ISPADMIN2:

```
net stop bio-monitor
net stop bio-cache
net stop bio-redis
```

If there are other BioTime or ZKTeco services, stop them also.

3. **Stop BioTime services on IBN-SERVER.** Open Command Prompt as Administrator on IBN-SERVER:

```
net stop bio-monitor
net stop bio-cache
net stop bio-redis
```

4. **Run Schema Compare.** In Visual Studio or SQL Compare:

Side	Database
Source	ISPADMIN2 live biotime database
Target	IBN-SERVER biotime database

Include these object types:

- Tables
- Views
- Stored procedures
- Functions
- Triggers
- Indexes
- Primary keys
- Foreign keys
- Default constraints
- Check constraints
- Schemas
- User-defined types
- Sequences, if any

Apply the schema update to IBN-SERVER.

5. **Run Data Compare.**

Side	Database
Source	ISPADMIN2 live biotime database
Target	IBN-SERVER biotime database

Copy/sync all BioTime tables, not only attendance tables. Include:

- Employees/personnel
- Attendance transactions
- Devices
- Schedules/shifts
- Holidays/leaves
- Departments/areas
- User accounts
- System settings/configuration
- Calculation/result tables

6. **Fix SQL user mapping on IBN-SERVER.**

```
USE biotime;
GO

ALTER USER ISPADmin WITH LOGIN = ISPADmin;
GO

ALTER ROLE db_owner ADD MEMBER ISPADmin;
GO
```

7. **Update statistics on IBN-SERVER.**

```
USE biotime;
GO

EXEC sp_updatestats;
GO
```

8. **Optional but recommended: check database integrity.**

```
USE master;
GO

DBCC CHECKDB ('biotime') WITH NO_INFOMSGS;
GO
```

No output means no detected errors.

9. **Clear Redis cache/queue on IBN-SERVER.** This prevents old queue data from affecting the newly migrated database.

```
cd /d C:\ZKBioTime\redis

ren appendonlydir appendonlydir_bak_migration
mkdir appendonlydir
```

If these files exist, rename them also:

```
ren appendonly.aof appendonly.aof.bak_migration
ren dump.rdb dump.rdb.bak_migration
```

10. **Start BioTime services on IBN-SERVER.**

```
net start bio-redis
net start bio-cache
net start bio-monitor
```

Confirm in Services:

```
bio-redis      Running
bio-cache      Running
bio-monitor    Running
```

11. **Test BioTime on IBN-SERVER.** Open `http://192.168.64.4:8081` and test:

- Login
- Employees open correctly
- Devices open correctly
- Attendance records exist
- Manual correction works
- Attendance recalculation works
- Reports generate correctly

12. **Redirect devices/users to IBN-SERVER.**

```
Server IP: 192.168.64.4
BioTime URL: http://192.168.64.4:8081
```

13. **Keep ISPADMIN2 stopped as rollback.** Keep the old PC unchanged for a few days, but do not allow it to receive new live attendance punches.

Final Rule

```
Stop old BioTime
Sync schema
Sync data
Fix SQL user mapping
Clear Redis queue on new server
Start new BioTime
Test
Redirect devices/users
```

If there is any problem after migration, point users/devices back to ISPADMIN2 only if IBN-SERVER has been stopped first. Avoid running both systems as live systems at the same time.